Customer Shopping Behavior Analysis

1. Project Overview

This report examines customer shopping behavior using transactional records from 3,900 purchases across multiple product categories. The main purpose is to identify spending trends, customer segments, product interests, and subscription-related behavior in order to support stronger business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase information (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior indicators (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 entries in the Review Rating column

3. Exploratory Data Analysis using Python

The analysis began with data cleaning and preparation in Python:

- **Data Loading:** The dataset was imported with pandas.
- **Initial Exploration:** df.info() was used to inspect the structure, while .describe() was used to review summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	Na
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	Na
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	Na
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	Na
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	Na
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	Na
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	Na

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Null values were identified, and missing entries in the Review Rating column were filled using the median rating for each product category.
- **Column Standardization**: Column names were converted into snake case to improve readability and consistency.
- **Feature Engineering:**
  - Added an age_group column by grouping customer ages into ranges.
  - Added a purchase_frequency_days column based on purchase timing data.
- **Data Consistency Check**: Confirmed that discount_applied and promo_code_used carried overlapping meaning; as a result, promo_code_used was removed.
- **Database Integration**: The cleaned DataFrame was connected to PostgreSQL and stored there for SQL-based analysis.

4. Data Analysis using SQL (Business Transactions)

Structured SQL analysis in PostgreSQL was carried out to answer important business questions:

1. **Revenue by Gender** – Measured the total revenue contributed by male and female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Located customers who used discounts and still spent more than the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Total rows: 839 Query complete 00:00:(

3. **Top 5 Products by Rating** – Identified the products with the highest average review scores.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase values between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Evaluated customer count, average spending, and total revenue by subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Determined the five products with the greatest share of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Grouped customers into New, Returning, and Loyal segments based on their purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Ranked the most frequently purchased items within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with more than five purchases were more likely to hold a subscription.

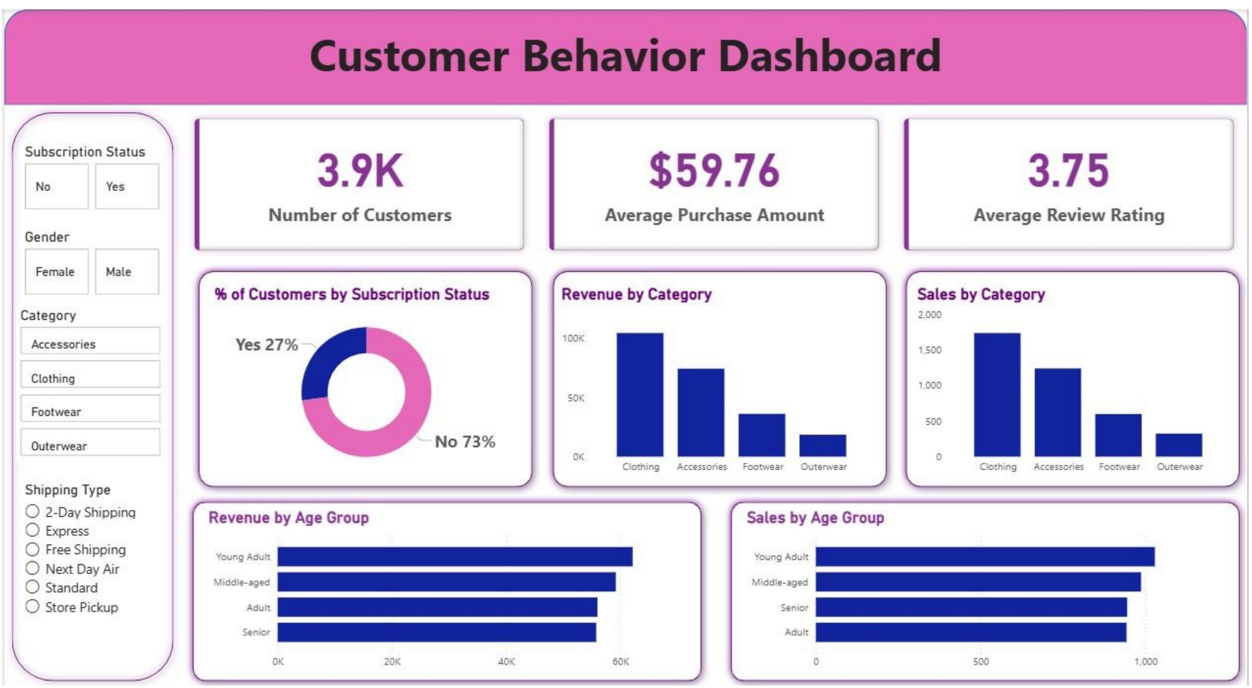
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated how much revenue each age group contributed overall.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

An interactive Power BI dashboard was created to present the findings visually and make the insights easier to explore.



6. Business Recommendations

- **Increase Subscriptions –** Offer stronger member-only benefits to encourage more customers to subscribe.
- **Customer Loyalty Programs –** Encourage repeat buyers with rewards so they can move into the Loyal segment.

- **Review Discount Policy –** Manage discount usage carefully to support sales without reducing profit too much.
- **Product Positioning –** Promote highly rated and best-selling products more prominently in campaigns.
- **Targeted Marketing –** Focus marketing efforts on the age groups that generate the most revenue and on customers who prefer express shipping.